

Enterprise AI Pilot Planning Assistant

End-of-Course Project Proposal | Chen Baiyi

1. Working title

Enterprise AI Pilot Planning Assistant: a structured tool that turns an SME use case into a reviewable AI pilot plan.

2. The problem, and why it matters

SME business leaders often want to trial an AI tool but do not know how to define a bounded pilot: who should test it, what tasks should happen first, what success looks like, and what feedback should be collected. Asking a general LLM directly produces uneven plans and forces users to write long prompts. The cost is slower, less comparable pilot decisions and a higher chance of testing the wrong use case. This project provides a first-draft plan only; a human project owner must review it before any pilot begins.

3. Who it is for, and the domain

The primary user is an SME operations or project manager who has been asked to assess one proposed AI use case. For example, Mei, an operations manager at a 40-person retailer, needs to decide whether five customer-service staff should trial an AI drafting tool for two weeks. After using the system, she should have a reviewable pilot plan with a named test group, staged tasks, a success metric, a feedback instrument, and explicit approval points. The first version is for SMEs and does not make recommendations for high-stakes medical, lending, hiring, legal, or security decisions.

4. Why AI - and which kind?

I will use a **single-turn foundation-model workflow**, not an agent, RAG system, or narrow ML model. The task is contextual structured writing: combine a small number of user-provided facts into a pilot-plan draft. A fixed document template cannot tailor recommendations to different industries, users, constraints, and pilot durations. The non-AI baseline is a rules-only template that inserts the same inputs into fixed headings. The AI system will use a structured web form and a fixed system prompt that returns: objective, test group, staged tasks, timeline, success metric, feedback instrument, risks, and human approval point. I will not use RAG because version 1 does not answer from a document collection. I will not use an agent because no tool use, memory, browsing, or multi-step autonomous loop is needed. This choice follows the Class 4 distinction between a prompt workflow and an agent, while keeping cost and failure modes proportionate to the task.

5. Proposed approach - build vs buy

I will use a hybrid build-and-buy architecture. The value I build is the structured input, validation, prompt, output schema, evaluation, and guardrails; the commodity language model is rented.

Layer	Decision	Rationale
Interface and input validation	Build: Streamlit + Python	Collects the minimum planning inputs and prevents incomplete requests.
Prompt orchestration and output schema	Build	Converts inputs into a controlled request and renders a reviewable plan.
Foundation model	Buy: OpenRouter API, initially openai/gpt-4.1-mini	No training or fine-tuning is justified for a four-week project.

Layer	Decision	Rationale
Retrieval/data layer	No retrieval in v1	The plan is based on the form inputs, not a proprietary document corpus.
Evaluation and observability	Build	Records synthetic case ID, output score, latency, token use, and estimated cost.

The smallest implementable path is: one SME scenario entered in a form -> one API call -> one structured plan. Version 1 excludes accounts, document upload, collaboration, integrations, and autonomous agents. I will report token-based estimated cost per plan and compare this cost with the usefulness of the output, reflecting the Class 5 focus on value, data readiness, operating model, and cost.

6. Data - this packs the intelligence

The application will not ingest a company database or retrieve confidential enterprise documents. Nevertheless, form inputs can be commercially sensitive, so users will be told not to enter personal data, client names, credentials, financial figures, contracts, or confidential strategy. Version 1 will not retain form responses after the session. The evaluation dataset will be a version-controlled synthetic CSV of **10 SME pilot scenarios**. Each row will contain case ID, industry, company size, AI use case, target user role, pain point, duration, constraint, and expected plan components. The cases will deliberately vary industry, ambiguity, data readiness, and risk level. A README will document how the cases were created. Stored evaluation records will contain only synthetic case ID, scores, latency, token usage, and estimated cost. This makes the evaluation reproducible without exposing real companies.

7. Success metric and how I will evaluate it

The primary metric is **pilot-plan completeness**. For each output, I will score five required elements as 0 or 1: (1) named test group, (2) dated or sequenced stages, (3) measurable success metric, (4) specified feedback instrument, and (5) human approval/checkpoint. The maximum score is 5 per case. I will run the same ten synthetic scenarios through (A) the structured form plus fixed system prompt and (B) the raw model with no form and no fixed prompt. My target is a mean completeness score of at least 4.0/5 for the structured system and at least 1.0 point higher than the raw-model baseline. A second reviewer will independently score at least three cases to identify rubric ambiguity. I will also report latency and estimated cost per plan.

8. Risks, limitations and responsible use

The principal risk is a plausible but unsuitable plan. Mitigation: every output is labelled “draft - human review required,” includes an approval checkpoint, and cannot be treated as an implementation decision. A silent failure occurs when the plan looks complete but proposes a test group, metric, or task that does not match the stated use case; I will detect this through the completeness rubric and manual review of the ten cases. Prompt injection or inappropriate user text is mitigated by delimiting user input, instructing the model to ignore embedded instructions, and restricting output to the defined schema. Privacy risk is mitigated by data minimisation, no persistent storage in v1, and an explicit ban on sensitive inputs. The tool is intended for low-risk internal pilot planning only; it must not be used to make automated decisions about people, health, credit, hiring, security, or legal matters. This applies the Class 6 focus on characteristic AI failure modes, review budget, accountability, and obligations to users.

9. Smallest first version

A user enters a fictional SME customer-service scenario, a target role, a two-week duration, and one constraint. The application validates the form, sends one controlled prompt to the rented model, and displays one structured draft plan. The version succeeds if, on a real synthetic test case, it generates all five required plan elements and a human reviewer can understand what to test, when, how success will be measured, and who must approve the next step. Everything else is deferred until this path works.